

Day 1: Introduction to Networking and the OSI Model Theory (1 hour): What is Networking? Overview of the OSI Model: Layers and their functions. Practical (1 hour): Use a network sniffer (e.g., Wireshark) to observe traffic and identify layers. Task: Create a diagram of the OSI Model with examples for each layer.

Task 1 :- Install Wireshark on your laptop, start a capture on your main network interface, and visit any website (like instagram.com). Stop the capture and note down the protocol names you see in the packet list.

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Here are the main protocols you see in Wireshark when visiting a website like Instagram :

- **DNS** : Finds the website's IP address (acts like a phonebook).
- **TCP** : Sets up a reliable connection between your device and the server.
- **TLS / SSL** : Encrypts your traffic so your data stays safe and secure.
- **HTTPS / HTTP** : Loads the actual website content (text, images, and videos).
- **QUIC** : A faster, modern protocol used by sites like Instagram and Google to speed up loading times.

Task 2 :- Pick any 3 packets from your Wireshark capture and identify which OSI layer each packet mainly represents (for example, HTTP for Application layer, TCP for Transport layer, etc).

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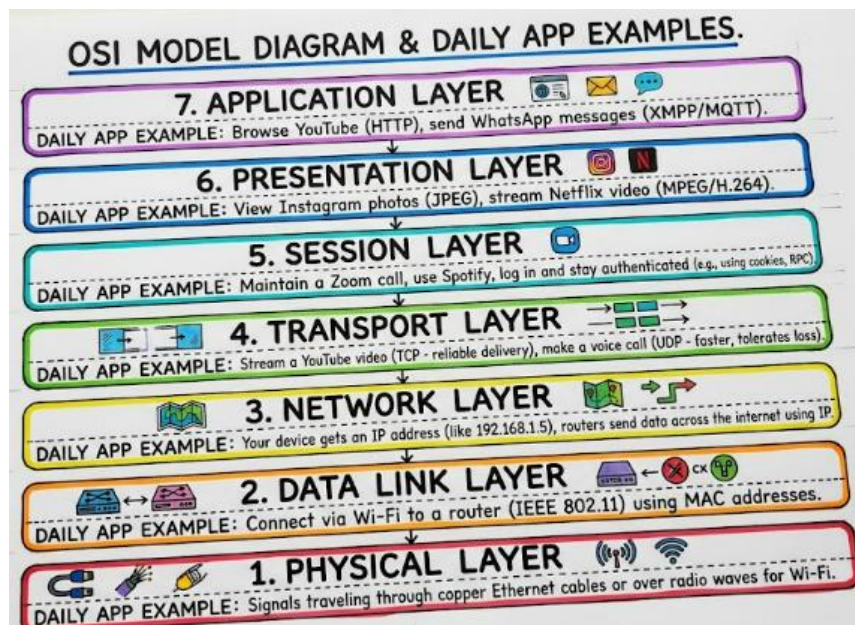
- **Packet 1 : HTTP / HTTPS :-**
 - OSI Layer: Application Layer (Layer 7)
 - What it does: Handles what you actually see and interact with on the web page.

- **Packet 2 : TCP :-**
 - OSI Layer: Transport Layer (Layer 4)
 - What it does: Delivers data reliably and makes sure no packets get lost.

- **Packet 3 : IP (IPv4 / IPv6) :-**
 - OSI Layer: Network Layer (Layer 3)
 - What it does: Addresses and routes the data to the correct destination on the internet.

Task 3 :- Create a colorful diagram of the OSI Model using any drawing tool (MS Paint, Canva, or even paper and upload a photo). For each layer, add a real-world example from apps you use daily (e.g., WhatsApp uses the Application layer, Instagram images use the Presentation layer, etc.).

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Task 4 :- Explain how sending a message on WhatsApp travels through at least 4 layers of the OSI Model, describing what happens at each layer in your own words.

ANS :-

1. Application Layer (Layer 7) :-

- What happens : You type "Hey!" and hit send. The app prepares your message using network protocols (like HTTPS or WebSocket) so it can be sent over the internet.

2. Transport Layer (Layer 4) :-

- What happens : Your message gets broken into smaller pieces called packets. It adds port numbers and chooses TCP to make sure the message arrives safely and in the right order without losing any words.

3. Network Layer (Layer 3) :-

- What happens : It adds destination IP addresses (like adding a postal address) so the internet routers know which path to take to send your message to WhatsApp's servers and then to your friend's phone.

4. Physical Layer (Layer 1) :-

- What happens : Your message is converted into raw binary signals—like radio waves for Wi-Fi/Cellular data or light signals in fiber-optic cables—and physically travels through the air/cables to reach the destination.